

AtliQ Technologies

Overview

AtliQ Technologies is a software company in Vadorada, India. It provides consultation services to help technological businesses upscale.

The human resources manager reached out with a dataset containing attendance information for each staff member within the company. The dataset was an Excel file and had four tables:

- Three tables containing daily attendance within 3 months
- One table which had the attendance key for the three other tables

	A	B	C	D	E	F
1	AtliQ		1 - Jun	2 - Jun	3 - Jun	4 - Jun
2	Employee Code	Name	Wed	Thu	Fri	Sat
3	Atq -406	Thanos Thakur	P	P	P	WO
4	Atq -462	Jarvis Singh	P	P	P	WO
5	Atq -411	Nevaeh Waller	P	P	P	WO
6	Atq -398	Bo Cordova	SL	P	P	WO
7	Atq -438	Mekhi Singleton	P	P	P	WO
8	Atq -366	Peter Pandey	P	P	P	WO
9	Atq -441	Loki Lal	P	P	P	WO
10	Atq -436	Phoenix Brady	P	P	P	WO
11	Atq -404	Ana Little	PL	P	P	WO
12	Atq -397	Grace Stone	P	P	P	WO
13	Atq -335	Tori Shannon	P	P	P	WO
14	Atq -443	Lyric Bartlett	P	P	P	WO
15	Atq -419	Kaylah Schultz	P	P	P	WO
16	Atq -467	Cason David	P	P	PL	WO
17	Atq -450	Miya Hampton	P	P	P	WO
18	Atq -405	Sofia Solis	P	P	P	WO
19	Atq -440	Mckayla Parker	P	P	P	WO
20	Atq -335	Madelynn Gray	P	WFH	ML	WO
21	Atq -343	Myla Klein	P	P	WFH	WO
22	Atq -334	Boston Morse	P	P	P	WO
23	Atq -446	April Ayers	P	P	P	WO
24	Atq -419	Faith Hamilton	P	P	P	WO
25	Atq -396	Joel Cruz	P	P	P	WO
26	Atq -459	Iris Woodard	P	P	P	WO
27	Atq -349	Trystan Ortega	P	LWP	LWP	WO
28	Atq -438	Miles Fields	P	P	P	WO
29	Atq -423	Rodrigo Price	P	P	P	WO
30	Atq -370	Ella Roth	P	HPL	P	WO
31	Atq -433	Jayce Fuentes	P	P	P	WO
32	Atq -375	Hailee Fitzpatrick	P	P	P	WO
June 2022 May 2022 Apr 2022 Attendance Key +						

Fig 1.1. Dataset preview in MS Excel

The dataset had **86 rows** and **45 columns** spanning **April - June 2022**.

Research Questions

The final deliverable needed was an automated dashboard that combined the three months and could update accordingly with newly added data. The following questions were to be answered via the dashboard:

1. **Working Preferences:** *if people prefer to work from home or work on-site*
2. **Attendance Information:** *which day is a majority of the staff present in the organization*
3. **Sick Leave Percentage:** *how many people are taking sick leaves, and how often do they take it*
4. **Leaves Percentage:** *how often are people going on leave, and how many of them are unforced*

Data Cleaning

I previewed the dataset on MS Excel; the first thing that really stood out was the structure. It was in a wide format with 45 rows. This would be a problem as it was human-readable but would be a tall order for the computer to read.

Each column had the attendance per day and then the totals for each attendance type. I also saw that the data had several missing values after June 19th and could potentially throw errors if included.

I loaded the dataset into power query and began the cleaning.

Given that the dashboard should update automatically for future data, the cleaning had to be automated, and this was a headache because each of the 46 columns had different header names across the three sheets.

After some googling, I combined the sheets into one (the fourth sheet was the attendance key and neededn't be combined, so I filtered it out).

Queries [4]		= Excel.Workbook(File.Contents("C:\Users\...	
CleanAndUnpivot [3]			
Template			
Transformations (Ap...			
CleanAndUnpivot			
Other Queries [1]			
Attendance_Cleaned			

	A ^B C Name	Data	A ^B C Item
	- Valid 100% - Error 0% - Empty 0%	- Valid 100% - Error 0% - Empty 0%	- Valid - Error - Empty
1	June 2022	Table	June 2022
2	May 2022	Table	May 2022
3	Apr 2022	Table	Apr 2022
4	Attendance Key	Table	Attendan

Fig 1.2. The single table containing all the sheets

I created a copy of the combined table and removed all other tables and columns besides the April column containing the attendance data. This was to serve as a template, where I cleaned up the table and stored the steps taken as a function that I would use to automate the same procedures on the other tables.

Queries [4]			= #"Removed Other Columns"{0}[Data]		
CleanAndUnpivot [3]			ABC 123 Column1		
Template			ABC 123 Column2		
Transformations (Ap...			Valid 8% Error 0% Empty 92%		
CleanAndUnpivot			Valid 8% Error 0% Empty 92%		
Other Queries [1]			Valid 8% Error 0% Empty 92%		
Attendance_Cleaned			Valid 8% Error 0% Empty 92%		
1	AtliQ		null		
2	Employee Code		Name		
3	Atq -406		Thanos Thakur		
4	Atq -462		Jarvis Singh		
5	Atq -411		Nevaeh Waller		
6	Atq -398		Bo Cordova		
7	Atq -438		Mekhi Singleton		
8	Atq -366		Peter Pandey		
9	Atq -441		Loki Lal		
10	Atq -436		Phoenix Brady		
11	Atq -404		Ana Little		
12	Atq -397		Grace Stone		
13	Atq -335		Tori Shannon		
14	Atq -443		Lyric Bartlett		
15	Atq -419		Kaylah Schultz		
16	Atq -467		Cason David		
17	Atq -450		Miya Hampton		
18	Atq -405		Sofia Solis		
19	Atq -440		Mckayla Parker		
20	Atq -335		Madelynn Gray		
21	Atq -313		Myla Klein		

Fig 1.3. April table before cleaning

- I removed the second row with irrelevant data and set the column headers appropriately.
- I unpivoted the columns with daily attendance information and corrected the data type.
- I renamed the column headers appropriately.
- I corrected the errors that were thrown up due to the total columns (they were inconsequential).

Queries [4]

CleanAndUnpivot [3]

Template

Transformations (Ap...

CleanAndUnpivot

Other Queries [1]

Attendance_Cleaned

✕

✓

fx

= Table.RemoveRowsWithErrors("#Changed Ty

	ABC 123 Employee Code		ABC 123 Name		Attr
	Valid Error Empty 100% 0% 0%		Valid Error Empty 100% 0% 0%		Valid Error Empty
1	Atq -406		Thanos Thakur		
2	Atq -406		Thanos Thakur		
3	Atq -406		Thanos Thakur		
4	Atq -406		Thanos Thakur		
5	Atq -406		Thanos Thakur		
6	Atq -406		Thanos Thakur		
7	Atq -406		Thanos Thakur		
8	Atq -406		Thanos Thakur		
9	Atq -406		Thanos Thakur		
10	Atq -406		Thanos Thakur		
11	Atq -406		Thanos Thakur		
12	Atq -406		Thanos Thakur		
13	Atq -406		Thanos Thakur		
14	Atq -406		Thanos Thakur		

Fig 1.4. April table after cleaning

Once the cleaning was done, I created a **parameter** based on all the steps taken. I inserted the parameter into the **Applied Steps** section of the **template table** by filtering the rows with it to serve as a test to see if it worked as expected.

As mentioned earlier, I then created a function from the *template* table with the parameter set to filter its rows.

I reverted to the main table, which contained all the attendance information, and created a **new column** by **invoking a custom function** (which was the function I had made).

Queries [4]		= Table.AddColumn(#"Filtered Rows", "Cleaned")	
CleanAndUnpivot [3]		Data	
Template		Item	
Transformations (Applied)			
CleanAndUnpivot			
Other Queries [1]			
Attendance_Cleaned			
	100%	Valid 100%	Valid
	0%	Error 0%	Error
	0%	Empty 0%	Empty
1	22	Table	June 2022
2	22	Table	May 2022
3	2	Table	Apr 2022

Fig 1.5. A new column (Cleaned) based on a custom function

I previewed all the newly created tables to check for errors and corrected a couple that resulted from the custom function.

I expanded the newly created column, formatted column names appropriately, removed unnecessary columns, and had a final combined table with unpivoted columns ready for analysis. I loaded it into Power B.I.

Queries [4]		= Table.SelectRows(#"Changed Type", each	
CleanAndUnpivot [3]			
Template			
Transformations (Ap...			
CleanAndUnpivot			
Other Queries [1]			
Attendance_Cleaned			

	A ^B _C Sheet Name	A ^B _C Employee Code	A ^B _C Name
	Valid Error Empty 100% 0% 0%	Valid Error Empty 100% 0% 0%	Valid Error Empty 100% 0% 0%
1	June 2022	Atq -406	Thanos
2	June 2022	Atq -406	Thanos
3	June 2022	Atq -406	Thanos
4	June 2022	Atq -406	Thanos
5	June 2022	Atq -406	Thanos
6	June 2022	Atq -406	Thanos
7	June 2022	Atq -406	Thanos
8	June 2022	Atq -406	Thanos
9	June 2022	Atq -406	Thanos
10	June 2022	Atq -406	Thanos
11	June 2022	Atq -406	Thanos
12	June 2022	Atq -406	Thanos
13	June 2022	Atq -406	Thanos
14	June 2022	Atq -406	Thanos

Fig 1.6. The final table for analysis

The Analysis

The attendance Key was integral to the analysis; the meaning of each key was defined as:

PL (Paid Leave), **SL (Sick Leave)**, **HPL (Half-day Paid Leave)**, **HSL (Half-day Sick Leave)**, **WFH (Work From Home)**, **FFL (Floating Festival Leave)**, **HFFL (Half-day Floating Festival Leave)**, **BL (Birthday Leave)**, **LWP (Leave Without Pay)**, **HLWP (Half-day Leave Without Pay)**, **BRL (Bereavement Leave)**, **HBRL (Half Bereavement Leave)**, **HWFH (Half Work From Home)**, **WO (Weekly Off)**, **HO (Holiday Off)**, **ML (Menstrual Leave)**, **HML (Half day Menstrual Leave)**.

I had to create measures to aggregate the data to answer the prevailing business questions. I created five (5) base measures:

- Total Working Days
- Total Present Days
- Total Remote Work Days

- Total Days on Leave
- Total Days on Sick Leave

✕

✓

```
1 TotalWorkingDays =  
2     VAR workdays = COUNT(Attendance_Cleaned[Value])  
3     VAR nonworkdays =  
4         CALCULATE(  
5             COUNT(Attendance_Cleaned[Value]),  
6             Attendance_Cleaned[Value] IN {"WO", "HO"}  
7         )  
8     RETURN  
9     workdays - nonworkdays
```

Attendance_Cleaned

...

Date

DayOfWeek

Employee Code

Name

OnLeave

Collapse ^

_Measures

...

Column

% OnLeave

% Presence

% SickLeave

% SickPeopleOnLeave

% WorkFromHome

RemoteWorkDays

TotalOnLeave

TotalPresentDays

TotalSickLeave

TotalWorkingDays

Collapse ^

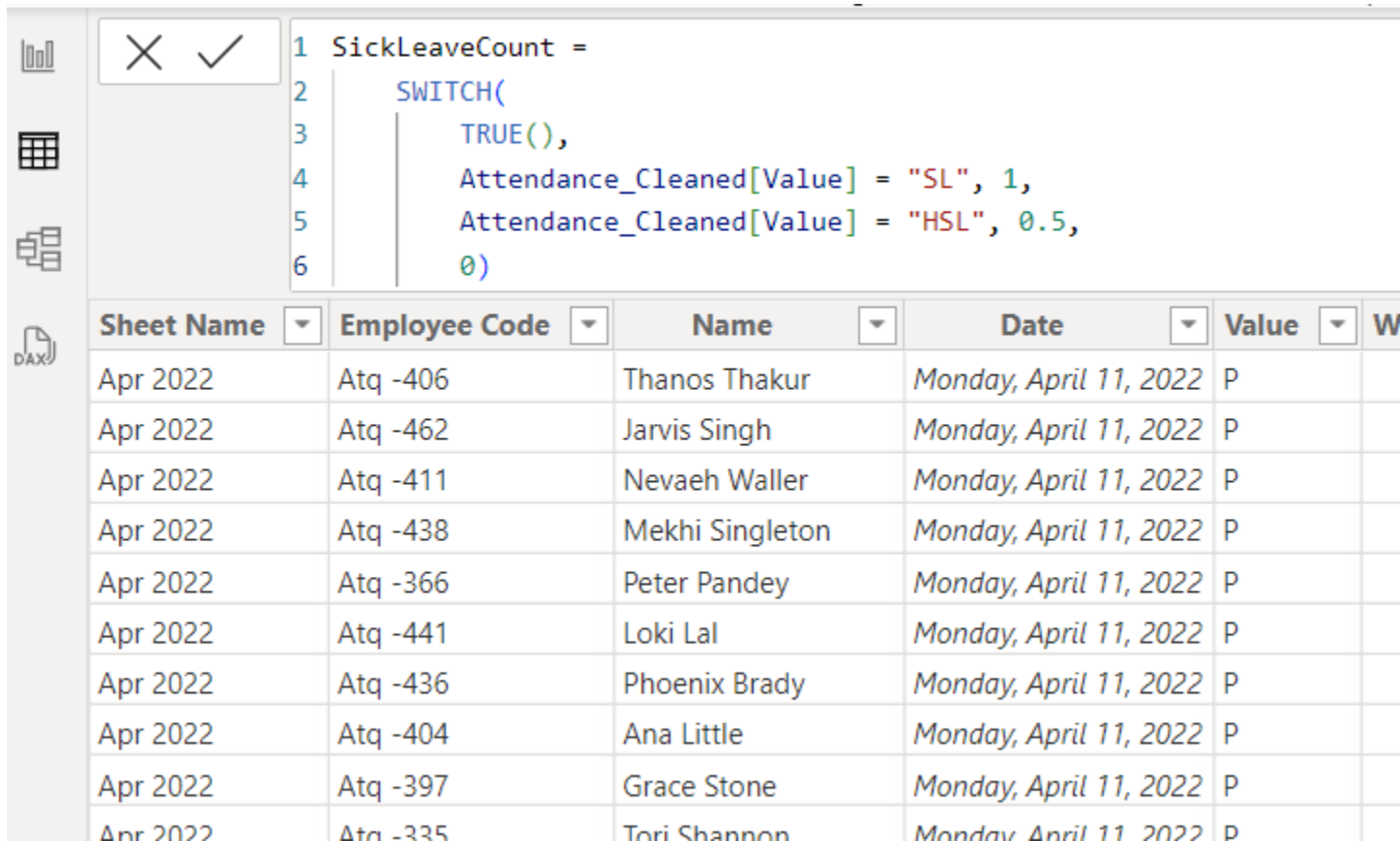
Fig 1.7. The table containing all created measures

I also created measures that calculated percentages based on the base measures. I created five (5) of them:

- *Percentage of Persons Present on a given day*
- *Percentage of Persons Working Remotely*
- *Percentage of Persons on Leave*
- *Percentage of Persons on Sick Leave*
- *Percentage of Sick Persons on Leave*

During the process of creating these measures, I also created calculated columns within the main table of data:

- *Work From Home Count*: based on the data, some people worked remotely for half a day, and this needed to be considered when calculating the number of persons present on a given day
- *Sick Leave Count*: accommodating those who took a half-day sick leave
- *On Leave*: aggregating everyone who took a type of leave on a given day
- *Day of Week*: based on the date column, the name of each working day



The screenshot shows a DAX editor interface. On the left, there are icons for a bar chart, a table, a document, and a DAX formula. The main area displays a DAX formula for a calculated column named 'SickLeaveCount'. The formula uses a SWITCH function to check the 'Attendance_Cleaned[Value]' for 'SL' (Sick Leave) and 'HSL' (Half Sick Leave), returning 1 and 0.5 respectively. Below the formula, a table is shown with columns: Sheet Name, Employee Code, Name, Date, Value, and W. The table contains 10 rows of data for April 11, 2022, with various employee codes and names, all having a 'Value' of 'P'.

```

1 SickLeaveCount =
2     SWITCH(
3         TRUE(),
4         Attendance_Cleaned[Value] = "SL", 1,
5         Attendance_Cleaned[Value] = "HSL", 0.5,
6         0)

```

Sheet Name	Employee Code	Name	Date	Value	W
Apr 2022	Atq -406	Thanos Thakur	Monday, April 11, 2022	P	
Apr 2022	Atq -462	Jarvis Singh	Monday, April 11, 2022	P	
Apr 2022	Atq -411	Nevaeh Waller	Monday, April 11, 2022	P	
Apr 2022	Atq -438	Mekhi Singleton	Monday, April 11, 2022	P	
Apr 2022	Atq -366	Peter Pandey	Monday, April 11, 2022	P	
Apr 2022	Atq -441	Loki Lal	Monday, April 11, 2022	P	
Apr 2022	Atq -436	Phoenix Brady	Monday, April 11, 2022	P	
Apr 2022	Atq -404	Ana Little	Monday, April 11, 2022	P	
Apr 2022	Atq -397	Grace Stone	Monday, April 11, 2022	P	
Apr 2022	Atq -335	Tori Shannon	Monday, April 11, 2022	P	

Fig 1.8. Calculated columns created to aggregate data

Visualizations

The first visual(s) I created were four (4) dynamic cards that displayed the percentages of those who were present, Worked Remotely, Were on Leave, or Were on Sick leave.

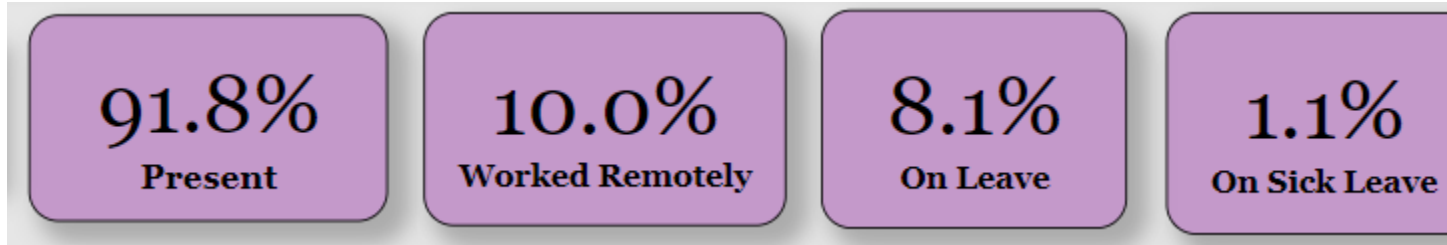


Fig 1.9. Dynamic cards

The cards, by default, showed the percentages for the three months. I then created a slicer to filter for each month.

The second visual was a table that displayed each member of staff's presence, percentage of time spent working remotely, and percentage of time spent on sick leave. This table was created to provide further information about each individual staff member's working preferences.

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Name	Present	Present Remotely	On Sick Leave
Aditya Walls	93.3%	7.1%	0.0%
Adriel Pace	98.2%	5.5%	0.0%
Adyson Moyer	98.2%	10.9%	0.0%
Alexander Davenport	100.0%	100.0%	0.0%
Alyson Huber	100.0%	0.0%	0.0%
Ana Little	76.4%	0.0%	1.8%
Andrew Cummings	95.2%	40.0%	0.0%
April Ayers	96.4%	0.0%	1.8%
Athena Rios	90.0%	7.4%	0.0%
Ayanna Atkins	58.9%	9.1%	10.7%

Fig 2.1. A table containing presence insights for each member of staff

The third visual was a treemap containing information about the number of people on leave per month per type of leave taken. This visual was created to answer the *Sick Leave Percentage* business question: to view the number of people who took leave, the kind of leave they took, and whether it was sick leave.

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STAFF ON LEAVE | (MONTH - LEAVE-TYPE)

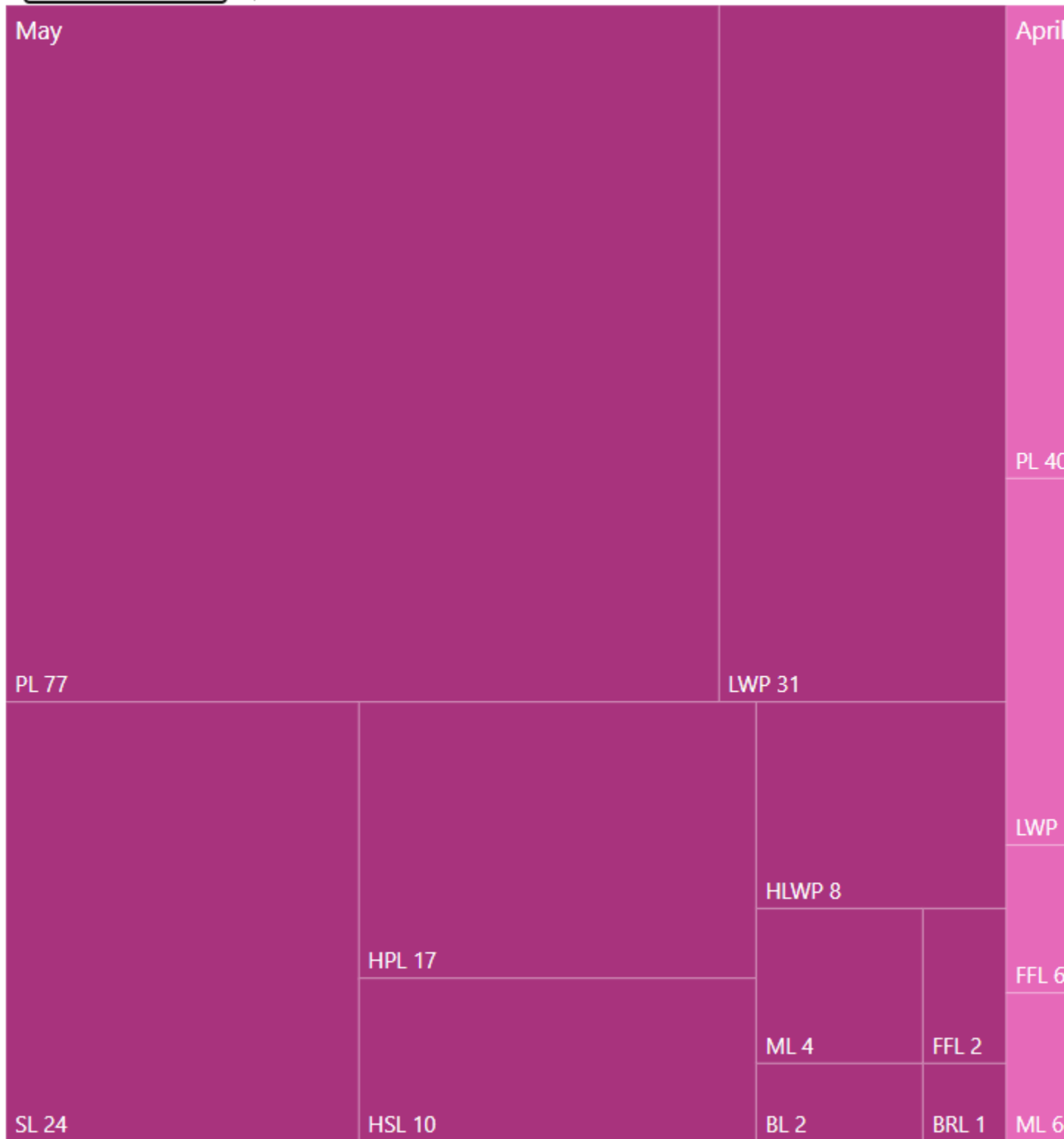


Fig 2.2. Treemap showing the number of people on leave per month per type of leave

The fourth visual(s) were three (3) stacked area charts showing the monthly trends for Attendance Information concerning staff presence, remote work, and sick leave. This visual provided information on how attendance changed, how much remote work had grown or fallen, and how often people took sick leave.



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STAFF PRESENCE | MONTH

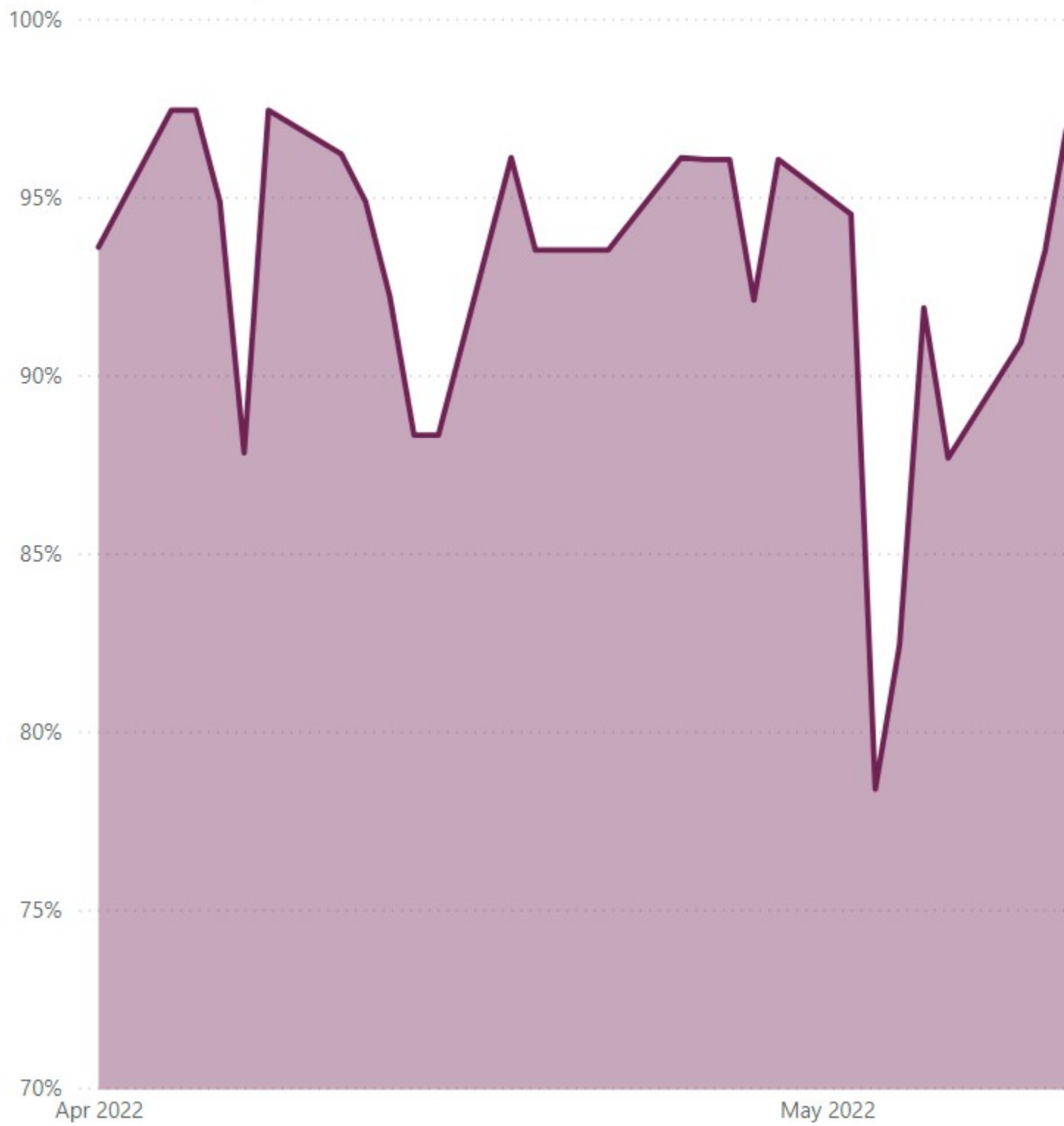


Fig 2.3. Stacked bar chart showing the staff presence trend over three months

The final visuals for the dashboard were three treemaps accompanied by the stacked area charts. These visuals showed the daily percentages of staff presence, remote work, and sick leave. They addressed the business question of Attendance Information by showing the days when people showed up the most.

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REMOTE WORK | DAY



Fig 2.4. Treemap showing the daily percentages of people working remotely

Insights and Conclusions

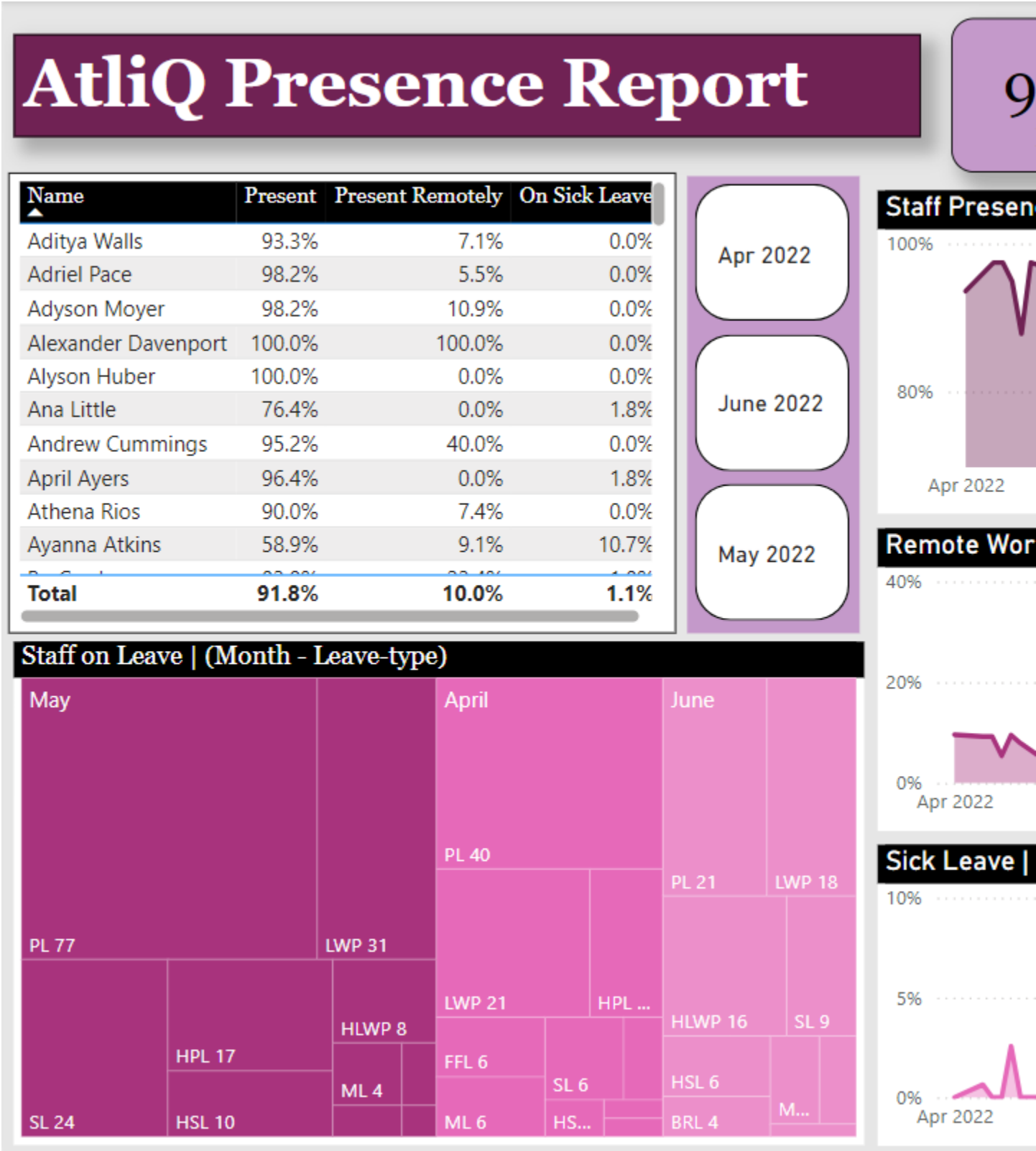


Fig 2.5. Final Dashboard

- 1. People are showing up regularly, and no drastic decisions regarding staff presence on regular working days are needed.**
- 2. Compared to the other two months, many people took leave in May, most of which were paid and unforced. It would be interesting to dig a little deeper into the external/internal events/circumstances during May that contributed to these decisions.**
- 3. Sick leave percentages were low and did not indicate anything to worry about.**
- 4. People preferred to work remotely on Thursdays and Fridays (no surprises there).**
- 5. On-site work was preferred over remotely, and people showed up the most on Mondays.**
- 6. Tuesdays in June had an average of 96% staff presence and should be the primary target day for any team-building activities or company-wide memos.**